

# A Variable-Rank Block-Power Criterion for Moving Quotient Tails

Bin Wang

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## Abstract

We replace the fixed-space convergence hypothesis of RH-269 by a direct variable-rank statement. The operators may act on changing quotient spaces, the selected rank and block length may grow, and only trace-norm/operator-norm power bounds are required. A geometric block decomposition then gives the full RH-246 logarithmic tail estimate beyond the moving head. The criterion is exact and conditional; no such uniform moving certificate is currently archived.

## 1 Hypotheses

Let  $C_\sigma$  act on a Hilbert space  $H_\sigma$ , let  $m_\sigma \geq 2$ , and assume uniformly

$$\|C_\sigma^{m_\sigma}\|_1 \leq K_\sigma, \quad \|C_\sigma^{m_\sigma}\| \leq \eta_\sigma < 1, \quad \|C_\sigma^r\| \leq L_\sigma^r \quad (0 \leq r < m_\sigma).$$

## 2 Theorem

**Theorem 2.1** (moving block tail). *For  $n = \ell m_\sigma + r$ ,  $0 \leq r < m_\sigma$ ,  $n \geq m_\sigma$ ,*

$$|\mathrm{Tr} C_\sigma^n| \leq K_\sigma \eta_\sigma^{\ell-1} L_\sigma^r.$$

*Consequently, if  $R > 0$  and  $\eta_\sigma R^{m_\sigma} < 1$ ,*

$$\sum_{n \geq m_\sigma} \frac{|\mathrm{Tr} C_\sigma^n| R^n}{n} \leq \frac{K_\sigma R^{m_\sigma}}{m_\sigma(1 - \eta_\sigma R^{m_\sigma})} \sum_{r=0}^{m_\sigma-1} (L_\sigma R)^r.$$

*Proof.* Factor  $C^n = (C^m)^{\ell-1} C^m C^r$ , use the ideal inequality for the trace, and group the geometric series by residue classes. The denominator  $n^{-1} \leq (\ell m)^{-1} \leq m^{-1}$  gives the displayed bound.  $\square$

## 3 Route consequence

If  $m_\sigma \rightarrow \infty$ ,  $\eta_\sigma R^{m_\sigma} \rightarrow 0$ , and

$$\frac{K_\sigma R^{m_\sigma}}{m_\sigma} \sum_{r=0}^{m_\sigma-1} (L_\sigma R)^r \rightarrow 0,$$

then, with a uniform prefix budget, the moving tail vanishes. A convenient stronger condition is

$$\limsup_{\sigma \downarrow 0} K_\sigma^{1/m_\sigma} R \max\{1, L_\sigma R\} < 1.$$

This is precisely the rank-growing replacement for a fixed limiting quotient. Supplying these bounds is an open operator problem; finite twelfth-power contractions do not supply them.